Fiscal Year:	FY 2004	**Task Last Updated:**	FY 12/02/2004
PI Name:	Hammond, Timothy M.D.		
Project Title:	Transcription factors mediating gene expression changes during renal cell rotating wall vessel culture.		
Division Name:	Physical Sciences		
Program/Discipline:	BIOTECHNOLOGY		
Program/Discipline--Element/Subdiscipline:	BIOTECHNOLOGY--Cellular biotechnology & tissue engineering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	timothy.hammond@duke.edu	**Fax:**	FY 9193238644
PI Organization Type:	NON-PROFIT	**Phone:**	9194285055
Organization Name:	Institute of Medical Research/Department of Veterans Affairs		
PI Address 1:	Institute For Medical Research, Inc.		
PI Address 2:	508 Fulton St, Building 15, Room 109		
PI Web Page:			
City:	Durham	**State:**	NC
Zip Code:	27705-3875	**Congressional District:**	1
Comments:	Alternate email: Timothy.Hammond@va.gov NOTE: PI previously at Tulane University until 2007 (Ed., 09/2010) Joint appointments Professor, Department of Internal Medicine, Nephrology Division, Duke University School of Medicine, and Staff Physician, Durham VA Health Care System.		
Project Type:	TRANSITIONAL (2004-2007)	**Solicitation / Funding Source:**	00-HEDS-03
Start Date:	08/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	1	**No. of PhD Degrees:**	2
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:	Kirby, Mary	**Contact Phone:**	
Contact Email:	mary.f.kirby@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	Hammond, Timothy		
COI Name (Institution):			
Grant/Contract No.:	NAG9-1344		
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Performance Goal Text:			
	Although there are several precedents for the transduction of mechanical forces into changes in gene expression, including heat shock proteins and shear stress responses, the molecular mechanisms by which mechanical culture conditions in the rotating wall vessel modulate gene expression in cultured cells remain largely unknown. Several characteristics of renal cells made they especially well suited for rotating wall vessel studies. First, reformation of microvilli, and expression of clinically important proteins such as P450 drug metabolism enzymes, and drug and hormone receptors provides a unique toxicology tool not available in any other form. Second, the in situ milieu of renal epithelial cells has shear levels similar to those in the rotating wall vessel. Last, several lines of evidence suggest defined transcription factors (from the vitamin D receptor, and TFII-I to TFII-B) translocated to the nucleus in a time dependent manner during initiation of rotating wall vessel culture. This application aims to utilize renal cells as a model		

Task Description:	system to identify and validate molecular mechanisms by which selected transcription factors control changes in specific clusters of genes during rotating wall vessel culture. Unifying hypothesis Nuclear translocation of specific transcription factors mediates changes in expression of specific gene clusters during rotating wall vessel culture of renal cells. Specific Aim I: To identify kinetic gene clusters responding to the “simulated microgravity” of rotating wall vessel mechanical culture conditions. Specific Aim II: To test whether blocking translocation of individual transcription factors specificity reverses the effects rotating wall vessel culture on select gene clusters. This aim will focus on the effects of up and down regulation of nuclear translocation of the vitamin D receptor nuclear translocation on gene expression clusters (Specific Aim IIa), as well as identifying other transcription factors for future studies (Specific Aim IIb). If successful, our work using micro-array technology will demonstrate specific clusters of genes, which change expression during the rotating wall vessel form of suspension culture, and some of the transcription factors, which mediate the changes. This will elucidate specific molecular mechanisms critical for optimization of suspension culture. This facilitates the production of bioproducts such as hormones from the kidney and other tissues, as well as new methods for testing drug interactions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work has two practical application to life on earth. The first application is to understand hoe to make cheap, clean bioproducts for use as biopharmaceuticals. This work is also designed to facilitate a quantum leap in studies of chemosensitivity by allowing individual drug regimens to be designed based on in vitro analysis of a patient diseased or normal tissue.
Task Progress:	Specific Aim I: To identify kinetic gene clusters responding to the “simulated microgravity” of rotating wall vessel mechanical culture conditions. Human renal proximal tubular cells were cultured for up to 8 hours in the rotating wall vessel and cell aliquots harvested for RNA extraction and analysis of gene expression using Affymetrix GeneChips. Publication of this data had been delayed due to patent issues with our industrial partner. These issues have now been resolved and the rotating wall vessel gene expression cluster manuscript rewritten, and we are in the process of submitting it for publication. At the protein level, we extracted nuclear protein from human renal epithelial cells grown in the rotating wall vessel and validated the purity and origin of the nuclear protein extracts by removal of surface membrane enzyme markers. Western blot analysis of this nuclear identified a group of more than 8 transcription factors, which translocate to the nucleus during initial cell culture in the rotating wall vessel. This data has been published in Gravitational and Space Biology Bulletin. Specific Aim II: To test whether blocking translocation of individual transcription factors specificity reverses the effects rotating wall vessel culture on select gene clusters To determine the effects of the vitamin D receptor, a transcription factor identified as one of the factors moving to the nucleus in Specific Aim 1, we tested for the ability to inhibit this translocation with vitamin D receptor tail peptides. Of three putative translocation sequences in the vitamin D preceptor, only 1 inhibited translocation, and was highly effective inhibiting >98% of the translocation. Human renal proximal tubular cells were cultured for up to 8 hours in the rotating wall vessel, with and without inhibition of vitamin D receptor translocation, and cell aliquots harvested for RNA extraction and analysis of gene expression using Affymetrix GeneChips. In this way we identified the vitamin D receptor dependent gene expression changes, which included protein kinase C changes and interleukin effects. Specific Aim III [Flight]: To identify kinetic gene clusters and transcription factor translocation in response to the true microgravity of space flight mechanical culture conditions. Human renal proximal tubular cells were cultured for up to 2 hours in the true microgravity of space on STS-106 and cell aliquots harvested for RNA extraction and analysis of gene expression using Affymetrix GeneChips. Publication of this data had been delayed due to patent issues with our industrial partner. These issues have now been resolved and the gene expression cluster manuscript rewritten, and we are in the process of submitting it for publication. This renal epithelial cell data has been compared for organ specificity using data sets from human liver cells flown in space by Stelsys LLC, to the whole organ kidney and liver effects by comparison to tissue gene expression in mice flown on STS-108 by Amgen, and mechanistically by comparison to a simple eukaryotic system in the yeast Saccharomyces cerevisiae. This has allowed comparison of organ specific changes, determination how good ground-based models are as predictors of microgravity effects, and prediction of transcription mediated drug metabolism changes in whole animals and astronauts in space. This application aims to utilize renal cells as a model system to identify and validate molecular mechanisms by which selected transcription factors control changes in specific clusters of genes during rotating wall vessel culture.
Bibliography Type:	Description: (Last Updated: 03/31/2026)
Articles in Peer-reviewed Journals	Nickerson CA, Ott CM, Wilson JW, Ramamurthy R, LeBlanc CL, Honer zu Bentrup K, Hammond T, Pierson DL. "Low-shear modeled microgravity: a global environmental regulatory signal affecting bacterial gene expression, physiology, and pathogenesis." J Microbiol Methods. 2003 Jul;54(1):1-11. Review. PMID: 12732416 , Jul-2003
Articles in Peer-reviewed Journals	Klassen RB, Crenshaw K, Kozyraki R, Verroust PJ, Tio L, Atrian S, Allen PL, Hammond TG. "Megalin mediates the renal uptake of heavy metal metallothionein complexes. " Am J Physiol Renal Physiol. 2004 Sep;287(3):F393-403. Epub 2004 May 4. PMID: 15126248 , Sep-2004
Articles in Peer-reviewed Journals	Cubano L, Allen PL, Stodieck L, Genova J, Klassen RB, Love J, Hammond TG. "Transcription factor translocation during changes in renal cell mechanical culture." Gravitational and Space Biology Bulletin. 2004 Jun;17(2):75-82. http://gravitationalandspaceresearch.org/index.php/journal/article/view/321 , Jun-2004

Patents	pending pending Feb-2004 Timothy G. Hammond and Patricia L Allen "Production of urokinase by immortalized human renal epithelial cells on beads"
Presentation	Hammond TG "Transcription factors mediating gene expression changes during" Experimental Biology 2004 Apr-2004
Presentation	Hammond TG "Transcription factors mediating gene expression changes during" American Society of Microbiology May-2004
Presentation	Hammond TG "Transcription factors mediating gene expression changes during" NASA Cell Science Conference Feb-2004